

# Patti Royale — an in-game commerce store

Product Associate assignment · PlaySuper

Karan Makol · karan.makol1@gmail.com · August 2026

Live prototype: <https://daenerysdrag.github.io/patti-royale-playsuper/>

Code: <https://github.com/DaenerysDrag/patti-royale-playsuper>

Best viewed on a laptop — the PM console docks on the right and the reasoning callouts sit beside the phone. Works on mobile too.

## Thesis

An in-game store that behaves like a shop will fail, because the player did not come to shop. There is no purchase intent to capture. The job is to convert accumulated play into a reason to play again.

In e-commerce, the cart is dead intent. In a game, the cart is a retention loop.

So I did not build a storefront. I built a coin sink that lifts retention without touching IAP.

## Assumptions

| WHAT I ASSUMED |                                                                                                                                            |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Game           | <i>Patti Royale</i> — casual trick-taking cards. 1M DAU, India, Android-first. 1.8 sessions/day, 4 matches/session. Baseline ARPDau ₹0.80  |
| Player         | Zero purchase intent on entry. ~40 seconds of attention between matches. Starts from the assumption that in-game "free rewards" are a scam |
| Archetypes     | <b>Grinder</b> (daily, never pays, 528 coins/day) · <b>Dipper</b> (2–3×/week, churn-risk, 61/day) · <b>Whale</b> (buys IAP, 235/day)       |
| Currency       | Win 50 · loss 15 · daily streak 40. Peg: <b>10 coins = ₹1</b>                                                                              |
| Non-goals      | Does not touch the existing cosmetic coin sink. Does not touch IAP pricing. Coins never buy in-game power                                  |
| Who pays       | 🔑 The <b>brand</b> funds the discount as customer-acquisition cost — not the studio                                                        |

### The consequence of that last row

If the brand funds the discount as CAC, then the constraint on this economy is **brand CAC budget and coupon inventory — not coin supply**.

Coins can be generous. What must be rationed is **conversion**. That is why there is a redemption cap (2/week) and no paywall on earning — and it is why I never modelled coins as a studio cost.

## Tools used

| TOOL                                                                | WHAT FOR                                                                                                                                                            |
|---------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Claude Code (Opus 5)                                                | The build — architecture, every screen, the trick-taking match logic, the event bus, the economy derivation, these docs                                             |
| Claude Design                                                       | A parallel visual pass: I designed the store as a multi-screen canvas, then ported its typography, button geometry and instrumentation panel into the working build |
| React 19 · TypeScript · Vite · Tailwind 4 · framer-motion · zustand | Stack. No backend; state persists to <code>localStorage</code>                                                                                                      |

| TOOL                   | WHAT FOR                                                                   |
|------------------------|----------------------------------------------------------------------------|
| GitHub Pages + Actions | Deploy. Every push to <code>main</code> typechecks, builds and republishes |

How I used them matters more than that I used them.

I wrote a **7-phase gated brief** for Claude Code first and made it stop at checkpoints for review, so the decisions stayed mine. Twice it flagged that my own numbers broke — the original earn rate made a "this month" reward affordable in five days, and a single set of price bands cannot be "reach today" for both a Grinder and a Dipper. Both flags changed the design. The economy here is the second version, not the first.

The Claude Design pass was deliberately **not** adopted wholesale, and what I rejected from it is the more useful thing to know. I took its type scale, its pill button geometry, and its instrumentation-panel treatment, which were all better than my first pass. I left behind its light iris-and-cyan palette — a white SaaS surface bolted to a dark felt card table would read as two products — and I left behind its economy readout panel, because that design's own footnote said its funnel and economy figures were "*illustrative, not instrumented*." This build's console can't carry a section it would have to fake.

The core question: how should this differ from e-commerce?

| E-COMMERCE ASSUMES                       | IN-GAME REALITY                                                                   | SO I BUILT                                                                       |
|------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| The user arrives <b>with intent</b>      | The player came to play; commerce interrupts                                      | The store opens at the <b>post-match moment</b> , never as a nav tab             |
| <b>Search</b> is the discovery primitive | 40-second window; choice paralysis kills it                                       | <b>No search, no filters.</b> 12 items, 3 fixed groups                           |
| Price is an <b>economic fact</b>         | Coins are printed by the studio and cost the player <i>time</i>                   | Price shown in <b>effort</b> — "3 more wins" — computed from your earn rate      |
| Not buying is <b>neutral</b>             | "You can't afford this" demotivates — damaging the retention this was sold to fix | <b>Zero dead ends.</b> Every locked item shows a ring and a Play button          |
| <b>Infinite catalog</b> wins             | Infinite catalog dilutes                                                          | 12 SKUs, digital only                                                            |
| A <b>cart</b> for multi-item baskets     | Single-item impulse only                                                          | No cart. Two taps                                                                |
| Currency is <b>fungible cash</b>         | A currency that buys real goods will cannibalise IAP                              | Real-world rewards only. <b>Never in-game power</b>                              |
| The store is <b>the destination</b>      | Every second in-store costs the studio the session                                | Exit CTA is " <b>Back to match</b> ", never "Continue shopping"                  |
| <b>Returns</b> handle regret             | You cannot refund someone's <i>time</i>                                           | <b>Coins refunded if a voucher expires unused.</b> No e-commerce analogue exists |
| Growth = <b>more traffic</b>             | Growth = <b>more reasons to play</b>                                              | <b>Goals</b> — below                                                             |

Goals — the mechanic the whole thing rests on

Reserve a reward and it becomes a **ring in the game HUD**, price-locked for 7 days. Every hand fills it. At 100% the game tells you.

This inverts the funnel from `browse → buy` into `want → play → afford → buy`. Commerce stops being an interruption and becomes the reason for the next session. It is also the only part of this design that plausibly moves D7 retention —

which is what a studio is actually buying.

Two deliberate consequences:

- `goal_created` is the conversion event that matters, not `payment_success`. Creating a goal costs the player nothing and costs the brand nothing, and it is what brings them back.
- **A player holding no goal gets one manufactured.** The reward screen surfaces the nearest out-of-reach item rather than a catalog: close enough to want, far enough to play for.

---

## The economy, and one derivation worth your time

---

Coin cap is a property of the **item**, never of the player: Tier 1 and 2 are fully coin-funded, **Tier 3 caps coins at 40% and the rest is cash** — PlaySuper's model, and the only place the coin↔cash slider appears. One flat rule, identical for everyone. I deliberately rejected loyalty tiers and engagement-based pricing: charging a more-engaged player more for the same item is manipulative, and it reads that way the moment two players compare screenshots.

**Back-solving your published +5.2% ARPDAU** rather than guessing a redemption rate:

```
₹0.80 baseline × 5.2% lift           = ₹0.0416 incremental per DAU per day
₹700 order × 10% commission × 50%   = ₹35 studio revenue per redemption

⇒ 0.0416 / 35 = 0.119% of DAU must redeem daily
⇒ ≈1,189 redemptions/day at 1M DAU ⇒ ≈₹5.35M/month of brand-funded discount
```

**Only about 1 player in 840 per day.** That is a tiny conversion requirement, and it changes the product: optimise for **retention breadth** — many players holding a goal — over **conversion depth**. A held goal costs the brand nothing and returns a player tomorrow; a checkout costs CAC and may not.

It is also why the conversion event I optimise for is **reserving a goal**, not completing a payment. Full funnel and event spec in `docs/03-event-taxonomy.md`.

(₹700 AOV, 10% commission, 50% studio share are my assumptions. If commission is 5%, the required redemption rate doubles and the catalog has to get cheaper — that's the first number I'd want from you.)

**One thing I decided not to fake.** A delivered coupon is not brand revenue — the brand's real conversion happens off-platform, days later, at the merchant's checkout. So the prototype's console shows the redemption it can observe and greys out the one it cannot, rather than inventing a figure. Closing that gap needs unique per-player codes and a brand-side redemption webhook. Until it exists, every ROI number quoted to a brand is a proxy and should be labelled as one.

---

## Three questions I'd want answered first

---

Each of these is an assumption I couldn't verify, and each one changes the design rather than just the copy.

1. **What coin-cap ceiling do brands actually accept?** I used 40% on Tier 3. It sets the whole economy and it is a guess.
2. **Who eats fulfilment failure — brand, PlaySuper, or the studio?** A dead coupon damages trust in *the game*, so the studio has the most to lose and the least control. That asymmetry needs an owner before it needs a process.
3. **What is the real commission and studio split?** My headline derivation — 0.12% of DAU — rests on ₹700 × 10% × 50%. At a 5% commission the required redemption rate doubles and this catalog is priced wrong.